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Motor unit activation strategy during a sustained isometric contraction of finger flexor muscles in elite climbers

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Abstract
The aim of the study was to evaluate, by an electromyographic (EMG) and mechanomyographic (MMG) combined approach, whether years of specific climbing activity induced neuromuscular changes towards performances related to a functional prevalence of fast resistant or fast fatigable motor units. For this purpose, after the maximum voluntary contraction (MVC) assessment, 11 elite climbers and 10 controls performed an exhaustive handgrip isometric effort at 80% MVC. Force, EMG and MMG signals were recorded from the finger flexor muscles during contraction. Time and frequency domain analysis of EMG and MMG signals was performed. In climbers: (i) MVC was higher (762 ± 34 vs 512 ± 57 N; effect size: 1.64; confidence interval: 0.65–2.63; $P < 0.05$); (ii) endurance time at 80% MVC was 43% longer (34.2 ± 3.7 vs 22.3 ± 1.5 s; effect size: 1.21; confidence interval: 0.28–2.14; $P < 0.05$); (iii) force accuracy and stability were greater during contraction ($P < 0.05$); (iv) EMG and MMG parameters were higher throughout the entire isometric effort ($P < 0.05$). Collectively, force, EMG and MMG combined analysis revealed that several years of specific climbing activity addressed the motor control system to adopt muscle activation strategies based on the functional prevalence of fast resistant motor units.

Keywords: *sport climbing, electromyogram, mechanomyogram, handgrip*

Introduction

Climbing is a unique sport activity, in that the role of the upper limbs and the predominantly vertical motion distinguishes it from all other land-based movements (Quaine & Martin, 1999). The action of climbing involves sustained isometric contractions of the forearm and finger muscles for most of the effort (Billat, Palleja, Charlaix, Rizzardo, & Janel, 1995). Static contractions are alternated to concentric movements of the upper and lower limbs, with partial resting periods between. The inability to generate and/or to sustain the finger flexor muscle force required to take the hold is the major reason for a fall or a movement failure (Watts, Daggett, Gallagher, & Wilkins, 2000), indicating that maximal isometric and concentric strength of the upper limb muscles are crucial factors in rock climbing (Espana-Romero et al., 2009; Grant, Hynes, Whittaker, & Aitchison, 1996; MacLeod et al., 2007; Quaine & Vigouroux, 2004).

Several years of climbing activity have been observed to induce both structural and functional changes in forearm muscles, such as muscle fibre hypertrophy (Watts, Newbury, & Sulentic, 1996), increased capillary density (Thompson, Farrow, Hunt, Lewis, & Ferguson, 2014), enhanced vasodilator capacity and oxygenation (Ferguson & Brown, 1997; Fryer et al., 2015) and a different motor unit activation strategy (Vigouroux & Quaine, 2006). These adaptations are similar to those induced by heavy-intensity isometric training (Rabita, Perot, & Lensel-Corbeil, 2000).

In general, the central nervous system coding of the motor unit activity (recruitment and firing rate) may be studied through the force oscillations (Taylor, Christou, & Enoka, 2003), and the electromyographic (EMG) (Basmajian & De Luca, 1985) and mechanomyographic (MMG) (Orizio, Gobbo, Diemont, Esposito, & Veicsteinas, 2003) parameters obtained during muscle action from signal time and frequency domain analysis. Therefore, new

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opportunities have arisen to study finger flexor muscle motor control characteristics in climbers. A few papers analysing the three signals during isometric contraction at different force intensities found significant differences in the motor unit activation strategy between climbers and controls. The results were compatible with the hypothesis of a shift of climbers' muscles towards a functional prevalence of faster motor units (Esposito et al., 2009; Koukoubis, Cooper, Glisson, Seaber, & Feagin, 1995; Watts, Joubert, Lish, Mast, & Wilkins, 2003). Whether the suggested shift is mainly dominated by the functional prevalence of the fast fatigable or fast resistant motor units is still an open question. Fast twitch motor units, indeed, can be classified as fast fatigable or fast resistant, according to their fatigability during a sustained muscle activation due to different structural, metabolic and functional characteristics (Schiaffino & Reggiani, 2011). Fast twitch motor units start to be recruited during isometric contraction beyond about 30% of the maximum voluntary contraction (MVC) and fully recruited at about 60–80% MVC, at least in larger muscles (Basmajian & De Luca, 1985). Then, the choice of 80% MVC force intensity in this study was supported by the fact that in this level of effort all motor units should be recruited and active (Basmajian & De Luca, 1985). Although climbing activities are known to require mostly intermittent contractions of relatively short duration, a sustained isometric contraction at a high intensity level under a controlled experimental condition may provide more insights into the possible prevalence of fast fatigable or fast resistant motor units throughout the muscle action. Indeed, given the higher fatigability of fast fatigable motor units during high intensity tasks (Fitts, 1994), an EMG, MMG and force combined analysis during a sustained contraction at 80% MVC (with a fully recruited motor unit pool) could shed more light on the open question.

On these bases, the aim of the study was to compare the electrical and mechanical response of the finger flexor muscles in elite climbers and sedentary controls during a sustained isometric contraction at 80% MVC. Should the years of climbing practice have induced functional prevalence of fast resistant motor units in finger flexor muscles, this would result in a higher maximum strength and endurance time, with different electromechanical behaviour through muscle action compared to matched controls. For this purpose, the electromechanical response of the finger flexor muscles was assessed during a sustained isometric contraction by an EMG, MMG and force combined approach.

Methods

Participants

Eleven elite climbers and ten healthy sedentary controls, with no previous upper limb injuries and without any recent history of practice with forearm contractions during sports or work, volunteered to participate in the study. Climbers were high-level athletes competing at the international level (mainly bouldering) for several years, with a maximum climbing ability level between 8a and 8b + , according to the European Scale of climbing difficulty evaluation. Therefore, they can be classified as élite athletes according to the climbing ability conversion table proposed by Draper et al. (2011). Their physical and anthropometric characteristics are given in Table I. Controls were healthy, physically active students of the local university.

All participants were right-handed males. All participants received detailed information on the purpose and the procedures of the study. They all signed a consent form prior to the beginning of the study, which was approved by the local Ethical Committee and had been performed in accordance

Table I. Anthropometric characteristics of participants (mean ± *s*).

	Controls	Climbers	ES	CI
Age (years)	24.6 ± 4.3	22.4 ± 3.0	−0.57	−1.45 to 0.30
Body mass (kg)	73.8 ± 6.0	63.2 ± 2.9*	−2.19	−3.28 to −1.11
Stature (m)	1.77 ± 0.05	1.71 ± 0.04	−1.28	−2.22 to −0.34
BMI ($kg \cdot m^{-2}$)	23.9 ± 1.5	20.04 ± 1.0*	−2.94	−4.17 to −1.70
Years of activity (years)	–	12 ± 1	–	–
Circumference (cm)	23.0 ± 0.8	27.87 ± 0.53*	6.96	4.69 to 9.23
Skin fold$_{ANT}$ (cm)	1.80 ± 0.19	0.65 ± 0.04*	−8.24	−10.88 to −5.61
Skin fold$_{POST}$ (cm)	2.16 ± 0.77	1.04 ± 0.05*	−2.02	−3.08 to −0.97
MBA (cm^2)	22.1 ± 1.5	28.9 ± 1.2*	4.83	3.14 to 6.53

Notes: Average values of age, body mass, stature, body mass index (BMI), forearm circumference, forearm anterior (skin fold$_{ANT}$) and posterior (skin fold$_{POST}$) skin folds, and finger flexors muscle-bone area (MBA) in controls and climbers. ES, effect size; CI, confidence interval. *$P < 0.05$ between groups.

with the principles of the 1964 Declaration of Helsinki.

Experimental design and procedures

Participants reported to the laboratory twice. During the first visit, participants were familiarised with the experimental setup and procedures, to be able to maintain the required output force constant. On a different day, they returned to the laboratory to perform three MVCs on a handgrip ergometer, each lasting 3 s, with 5 min of rest between. The highest value was considered to be closest to the MVC. Then, participants were required to perform an isometric effort at 80% MVC to the point of fatigue, i.e. when they were no longer able to keep the force level constant within ± 5% of the target. Participants were verbally encouraged by the investigators to maintain the requested effort as long as possible.

Apparatus

A schematic drawing of the positioning of the participant on the ergometer and of the probe on the investigated muscle is given in Figure 1.

During all tests, participants were sitting on a chair with the elbow at 90° and positioned on the ergometer. The EMG, MMG and force signals were recorded during isometric contractions of the extrinsic finger flexor muscles of the dominant arm. Indeed, a study analysing EMG in climbers from *interosseous, brachioradialis, biceps brachii* and *flexor digitorum superficialis* muscles evidenced that the latter had the highest EMG activity during hanging (Koukoubis et al., 1995). As drawn in Figure 1, handgrip force was measured by a specifically designed handgrip ergometer, consisting of two cylinders, one fixed to the frame of the ergometer, and the other one adjustable and connected to the fingers and to a load cell (Interface, SM-2000 N, Scottsdale, USA; linear from 0 to 2000 N) (Esposito et al., 2009; Orizio et al., 1997). The distance between the two cylinders was individually chosen to provide the most forceful hand grip (Boadella, Kuijer, Sluiter, & Frings-Dresen, 2005). The hand was positioned with the first (fixed) cylinder in the cavity between the thumb and the index finger, and the second (adjustable) cylinder was positioned and blocked at the level of the second fingers phalanges. Each participant held the ergometer with the forearm kept in a position midway between pronation and supination, with the elbow blocked on a fixed support.

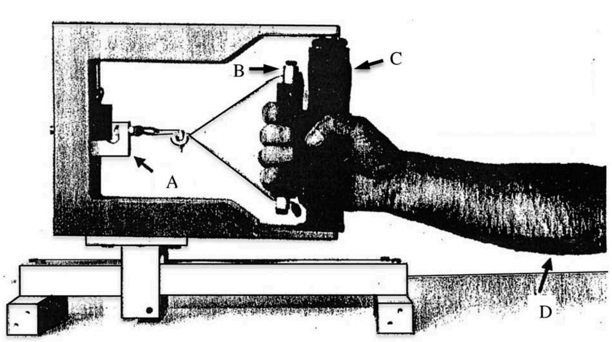

Figure 1. Schematic drawing of the ergometer. A, load cell; B, adjustable cylinder; C, fixed cylinder and D, EMG-MMG integrated probe position.

The force signal was displayed on a personal computer screen together with the target force to provide visual feedback. The EMG and MMG signals were detected by a specifically designed composite probe. The MMG records and quantifies the low-frequency transverse oscillations of active muscle fibres propagating to the skin surface (Orizio et al., 2003), thus reflecting the mechanical counterpart of the muscle electrical activity detected as surface EMG (Gordon & Holbourn, 1948). The integrated probe consisted of 1 cm spaced pair of silver bar electrodes (1 mm diameter; 1 cm long) and a piezoelectric contact sensor transducer (HP 21050A), with the tip placed between the two electrodes. The probe was positioned over the belly of *flexor digitorum superficialis* muscles after gentle abrasion of the skin with fine sand paper and cleaning with ethyl alcohol to achieve interelectrode impedance below 2000 Ω.

The two signals were amplified and filtered (EMG: 10–512 Hz; MMG: 2–128 Hz; force:). After analogue-to-digital conversion (RTI-815, Analog Device, Manning, SC, USA), EMG, MMG and force signals were stored on the hard disk of a personal computer with a sampling frequency of 1024, 512 and 128 Hz, respectively. The probe was then firmly secured to the muscle belly by an elastic band generating a pressure which did not induce changes in the frequency content of the piezoelectric contact sensor (Orizio, Perini, Diemont, & Veicsteinas, 1992).

Data analysis

Muscle-bone area was anthropometrically determined from the forearm circumference and the thickness of two skinfolds (De Koning, Binkhorst, Kauer, & Thijssen, 1986). The anterior and posterior skin fold thickness and the circumference of the investigated forearm were measured using a skin fold calliper (Holtain Ltd, Crymych, Wales) and a flexible ruler at the belly of *flexor digitorum superficialis* muscles. Compared to circumference alone, muscle-bone area determination permits to better estimate the functional transverse area of the investigated muscle group (De Koning et al., 1986).

To avoid interference of transient phenomena, only the middle 2 s of the MVC test, where the force signal was steady, were analysed. During the 80% MVC test up to exhaustion, EMG, MMG and force signal analysis was performed during the first 20 s of contraction within the target and during the last 2 s of contraction. Variables were then averaged every 2 s.

The coefficient of variation (CV) was estimated from the force signal as the ratio between the standard deviation of the force average value and the average value in the considered time window (CV = *s*/mean). CV can be considered as an index of muscle contraction stability. The distance of the force signal from the required target (ΔF) was also calculated and considered to be an index of muscle contraction accuracy.

The EMG and MMG signals were analysed in time and frequency domains, and the root mean square (RMS) and the power spectrum density distribution were then determined (custom built software). The spectra of the two signals were obtained using the maximal entropy spectrum estimation method (Orizio et al., 1992). From each spectrum, the mean frequency (MF) was then obtained. EMG and MMG average values during sustained isometric contraction have also been normalised to their initial value, considered to be 100%.

Statistics

Statistical analysis was performed using commercially available statistical software (SigmaStat, Systat Software Inc., USA). A Student *t*-test was performed to compare MVC and anthropometric variables between the two investigated groups. A two-way [time × group (climbers vs controls)] ANOVA for repeated measures for the first (force) factor was used for comparisons of EMG, MMG and force parameters throughout the sustained contraction. When a significant difference was retrieved, *post hoc* Holm-Sidak method was applied. The magnitude of the changes was assessed using effect size statistics with the standard error of effect size estimate and the lower and upper 95% confidence intervals. The statistical level of significance was fixed at an α level <0.05. Unless otherwise stated, here the results are expressed as mean ± standard error of the mean.

Results

Force

As expected, absolute MVC was significantly higher in climbers (762 ± 34 N) than in controls (512 ± 57 N; effect size: 1.64; confidence interval: 0.65–2.63; $P < 0.05$). When normalised per unit of the muscle-bone area, force values were still significantly different between the two investigated groups (26.5 ± 1.5 and 21.4 ± 2.2 $N \cdot cm^{-2}$ for climbers and controls, respectively; effect size: 0.83; confidence interval: −0.06–1.72; $P < 0.05$). Interestingly, the endurance time at 80% MVC was 43% longer in climbers (34.2 ± 3.7 s) than in controls (22.3 ± 1.5 s; effect size: 1.21; confidence interval: 0.28–2.14; $P < 0.05$).

For a more precise assessment of the differences between the two groups in terms of electromechanical response, we decided to compare only the first 20 s of the isometric task, i.e. the minimum amount of time reached by all participants, and the last 2 s of contraction immediately before exhaustion.

CV and ΔF values for climbers and controls during 80% MVC are shown in Figure 2. Climbers showed significantly lower CV values compared to controls throughout the entire period of contraction considered for the analysis (first 20 s). Climbers

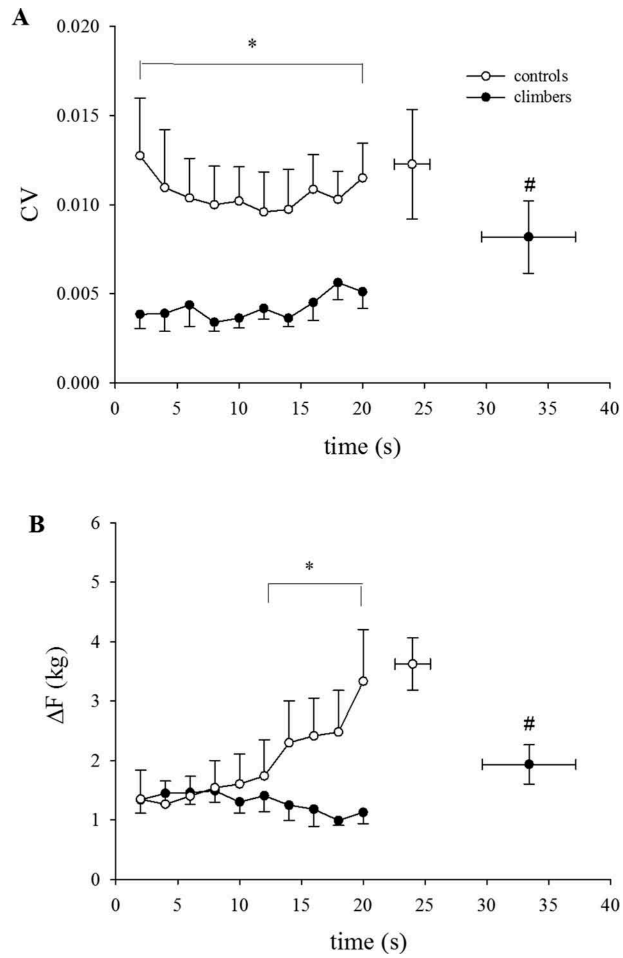

Figure 2. The coefficient of variation (CV) of the force (F) signal (Panel A) and the ΔF from F target (Panel B) during sustained contraction at 80% MVC, in climbers (●) and controls (○). $^*P < 0.05$ vs controls.

showed also significantly lower ΔF values compared to controls. Climbers were able to maintain the force output near to the required target the force signal throughout the entire task, while ΔF values in controls increased after the first 10 s of the test.

EMG

Figures 3 and 4 show the mean absolute and normalised values of EMG RMS and EMG MF as a function of time, during the first 20 s of 80% MVC contraction. In both groups, absolute EMG RMS increased significantly during the first 10 s of contraction, then stabilising at higher values afterwards. EMG RMS values were always significantly higher ($P < 0.05$) in climbers than in controls. EMG MF decreased significantly during the first 8 s of contraction, with no statistical differences between groups.

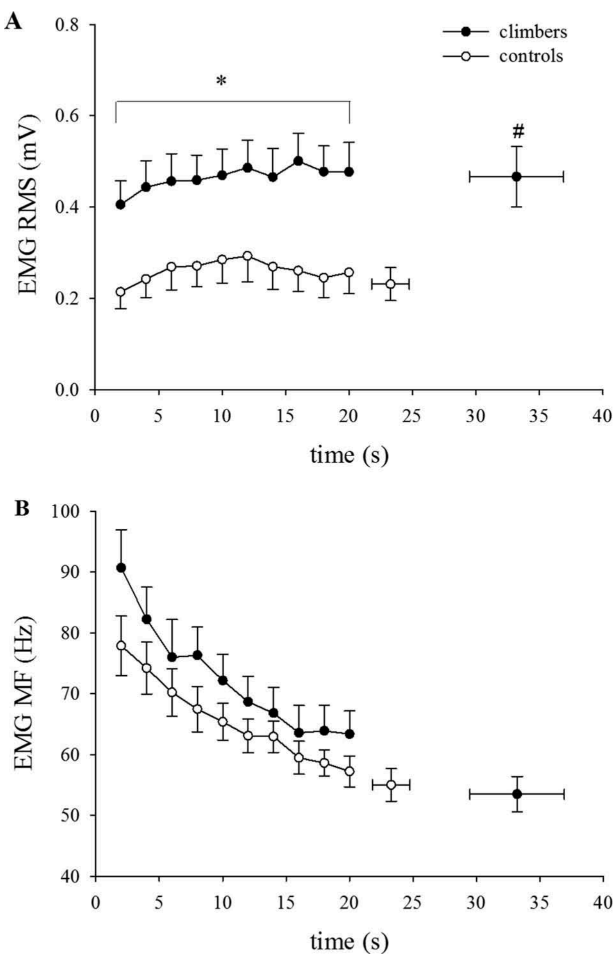

Figure 3. Mean values of EMG RMS and MF as a function of contraction time at 80% MVC, during the first 20 s and at exhaustion, in climbers (●) and controls (○). *$P < 0.05$ vs controls.

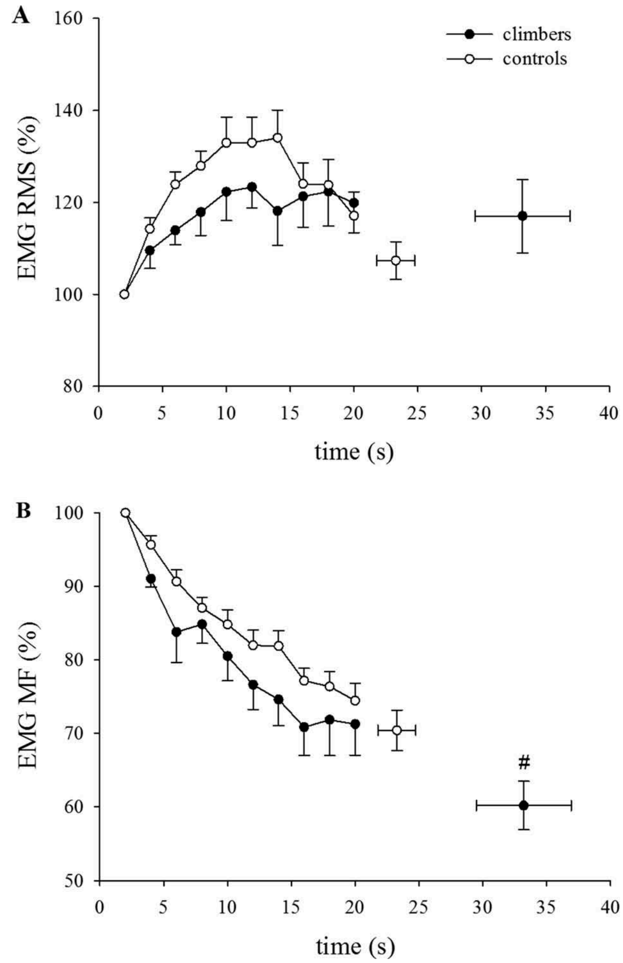

Figure 4. Mean values of EMG RMS and MF, normalised to the initial value (100%), as a function of contraction time at 80% MVC, during the first 20 s and at exhaustion, in climbers (●) and controls (○). *$P < 0.05$ vs controls.

MMG

In Figures 5 and 6, the mean absolute and normalised values of MMG RMS and MMG MF are plotted as a function of time. The absolute MMG RMS values were always significantly higher in climbers than in controls ($P < 0.05$). After an initial reduction during the first seconds of the task ($P < 0.05$ from second 2 to second 4), MMG RMS remained steady in both groups.

The absolute MMG MF values were always significantly higher in climbers than in controls ($P < 0.05$). MMG MF increased significantly at the beginning of the test from second 2 to second 6 in both groups ($P < 0.05$). Thereafter, MMG MF decreased after the first 6 s of the task in both groups ($P < 0.05$), achieving a steady-state condition during the rest of the contraction.

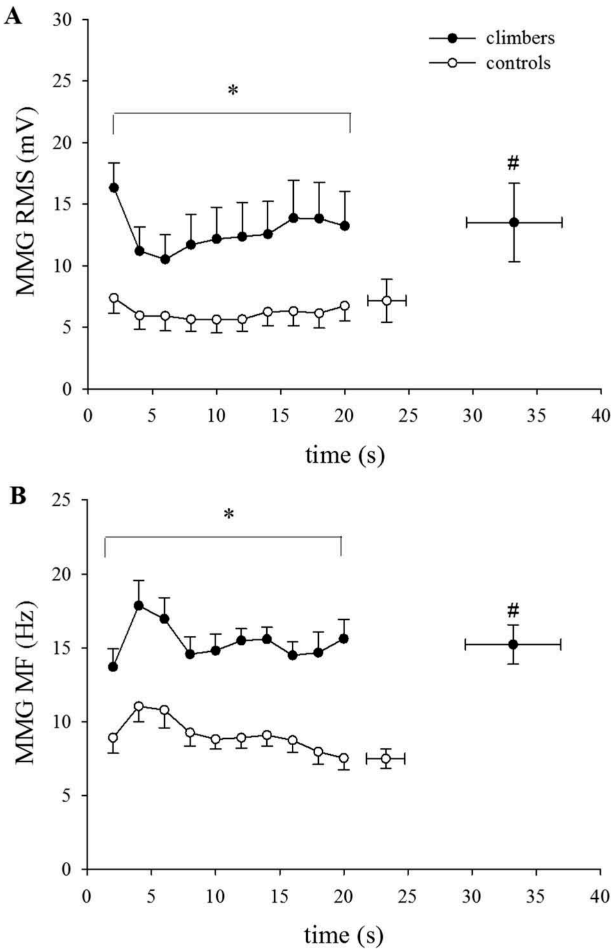

Figure 5. Mean values of MMG RMS and MF as a function of contraction time at 80% MVC, during the first 20 s and at exhaustion, in climbers (•) and controls (○). *P < 0.05 vs controls.

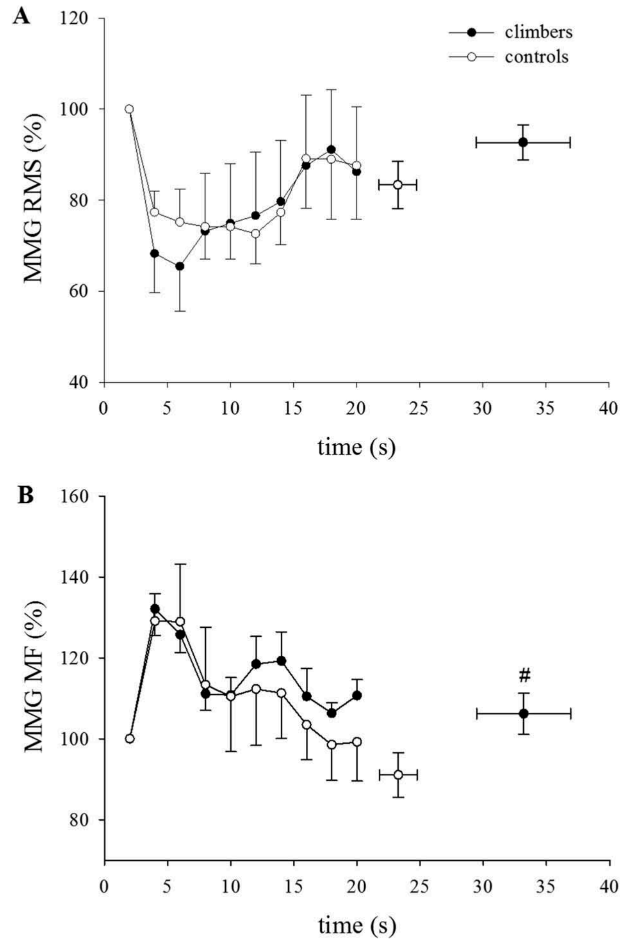

Figure 6. Mean values of MMG RMS and MF, normalised to the initial value (100%), as a function of contraction time at 80% MVC, during the first 20 s and at exhaustion, in climbers (•) and controls (○). *P < 0.05 vs controls.

Discussion

The main finding of the present study was that during isometric contraction at 80% MVC, climbers were able to sustain the force within the target for a longer period. Force stability and accuracy were also better maintained in climbers. In addition, EMG and MMG parameter dynamics during contraction provided further support to the hypothesis that several years of high intensity isometric activity induced a functional prevalence of type II fast resistant motor units in climbers.

Anthropometric and force parameters

Anthropometric characteristics. Several studies analysed climbers' anthropometric characteristics, which can result from either genetic heritage and/or training adaptations. Among other factors, muscle hypertrophy (Ogasawara, Yasuda, Ishii, & Abe, 2013; Yasuda, Loenneke, Thiebaud, & Abe, 2012) and maximum strength capacity (Macdonald & Callender, 2011; Sherk, Bemben, & Bemben, 2010; Tomaszewski, Gajewski, & Lewandowska, 2011) affect climbing performance. Our data about forearm circumference of climbers, often utilised as an index of hypertrophy, are in agreement with the values already present in the literature (Esposito et al., 2009; MacLeod et al., 2007; Tomaszewski et al., 2011), and suggest a marked hypertrophy in climbers, as witnessed also by previous reports on muscle-bone area or muscle cross-sectional area in climbers (Esposito et al., 2009; Sherk et al., 2010).

MVC. The higher MVC values of climbers in the present study compared to data on climbers previously reported (Giles, Rhodes, & Taunton, 2006) may be explained considering that we used an extremely adaptable handgrip ergometer, allowing participants to optimise their strength output. Moreover, athletes in the present investigation were elite climbers practising the bouldering discipline, which requires contractions of very high intensity

(Fanchini, Violette, Impellizzeri, & Maffiuletti, 2013). Interestingly, MVC and muscle-bone area were larger in climbers than in controls by +49% and +31%, respectively. Therefore, out of hypertrophy, also other factors could have played a role in the better performance of climbers, such as a higher number of active motor units and/or differences in the motor unit activation strategy (Esposito et al., 2009).

Coefficient of variation and ΔF. CV (an index of contraction stability) was significantly lower in climbers throughout contraction. The amplitude of force fluctuation, indeed, has been shown to be influenced directly by alterations in synergistic muscle activation (Shinohara, Yoshitake, & Kouzaki, 2009), by improvements in agonist–antagonist coactivation and/or discharge rate variability (Semmler, Steege, Kornatz, & Enoka, 2000) and to be inversely related to the number of the recruited motor units (Hamilton, Jones, & Wolpert, 2004). On these bases, years of isometric activity may have allowed climbers to involve a large number of usually non-recruitable motor units, with reduction in their relative tension ripple, leading to a greater ability to control and finally adjust the force output during exertion.

Δ*F* (an index of contraction accuracy) showed different values after 8 s of contraction when climbers, contrary to controls, were able to maintain the requested output tension. This may suggest an earlier onset of fatigue in controls and a less important functional role of fast resistant motor units. No other studies investigated these force parameters in climbers, thus comparisons cannot be made.

Endurance time

Several studies reported muscular endurance capacity in climbers, during both intermittent and continuous fatiguing contractions (España-Romero et al., 2009; Ferguson & Brown, 1997; MacLeod et al., 2007; Philippe, Wegst, Muller, Raschner, & Burtscher, 2012; Quaine, Vigouroux, & Martin, 2003a; Vigouroux & Quaine, 2006; Watts et al., 1996). While during intermittent contractions (España-Romero et al., 2009; MacLeod et al., 2007; Philippe et al., 2012; Quaine et al., 2003a; Vigouroux & Quaine, 2006) climbers always showed higher endurance capacity than controls, conversely, during continuous contractions (Ferguson & Brown, 1997; MacLeod et al., 2007; Philippe et al., 2012; Watts et al., 1996) no differences were retrieved between the two groups (Ferguson & Brown, 1997; Philippe et al., 2012). However, previous studies under continuous static effort were conducted mainly at about 40% MVC (Ferguson & Brown, 1997; MacLeod et al., 2007; Philippe et al., 2012), a force intensity markedly lower than that in our study. Only Watts et al. (1996) measured continuous isometric endurance time in high-level climbers at 70% MVC with findings similar to ours (about 35 s). In their study, though, no controls were investigated.

The longer endurance time in climbers should be supported by a wider energy availability for contracting fibres. The high intramuscular pressure during high-level static efforts may impair muscle blood flow (Sejersted et al., 1984). However, at high levels of effort, muscle blood flow is not as impaired as at 50% MVC (Kahn et al., 1998), suggesting that at 80% MVC, as in our study, muscle blood flow may not be the performance crucial limiting step. Climbers' forearm muscles present typical morphological and functional adaptations to isometric training, such as an increased capillarity (Jensen, Bangsbo, & Hellsten, 2004; Robbins et al., 2009). The presence of blood supply and the more efficient capillarisation providing better oxygenation also when performing isometric fatiguing contractions (Fryer et al., 2015; Usaj, 2002), together support the hypothesis that the higher endurance capacity of climbers may be related to the possibility of exploiting the fast resistant motor unit aerobic capacity, and consequently their possible main functional role in this kind of task.

EMG and MMG parameters

Only a few studies on finger flexor muscle strength in climbers reported also EMG data (Esposito et al., 2009; Koukoubis et al., 1995; Quaine, Vigouroux, & Martin, 2003b; Vigouroux & Quaine, 2006), and only one involved also MMG signal analysis (Esposito et al., 2009).

In previous studies, climbers with higher maximum force and endurance capacity presented also a lower decrease in EMG frequency content throughout the fatiguing effort (Quaine et al., 2003b; Vigouroux & Quaine, 2006). As a matter of fact, EMG frequency parameters are strongly influenced by metabolite accumulation within muscle fibres (Orizio, 2000). In particular, a decrease in pH with fatigue would lead to a higher rate of EMG median or mean frequency decay. Therefore, the reported lower decrease in EMG frequency content in time may be explained by climbers' previously mentioned greater muscular perfusion and oxygenation.

In the present study, EMG RMS increased during the first part of the task in both groups, then stabilising afterwards, despite the fact that climbers presented significantly higher values. According to Linssen et al. (1993), this EMG RMS behaviour may be related to the recruitment of additional motor units and or to an increase in their global firing rate (Maton & Gamet, 1989) or possible grouping of motor unit activity (Krogh-Lund & Jorgensen, 1993). However, while in controls, the rise in EMG RMS before the plateau lasted about 12 s, in climbers the increase in EMG amplitude continued up to about the 16th s. This suggests a greater ability to recruit motor units with higher firing rates for a longer period, and a better motor unit grouping capacity in climbers. The overall higher EMG RMS values in climbers can be explained by muscle fibre hypertrophy, determining an electrical source of larger size, and by the global firing increment, synchronisation and grouping of the active motor units (Esposito, Ce, Gobbo, Veicsteinas, & Orizio, 2005; Orizio et al., 2003). Moreover, the lower skinfold thickness in climbers may have contributed to the greater amplitude of the electrical signal in these athletes, due to a lower electrical filtering effect (Stegeman, Blok, Hermens, & Roeleveld, 2000). EMG MF showed a progressive decline in both groups. Climbers, though, revealed a tendency to higher frequencies throughout the entire isometric task, providing further support to the hypothesis that training-induced muscle hypertrophy brought to a shift to a functional prevalence of motor units with a higher conduction velocity.

The MMG RMS increases with motor unit recruitment, if the global firing rate is not so high to impair the fibre dimensional changes, which is the basic phenomenon generating the surface MMG (Orizio 1993; Orizio et al., 2003). The higher MMG RMS values in climbers are suggestive of a greater contribution to the MMG signal of larger motor units with a higher firing rate throughout the entire task. MMG MF, which is strongly related to the global motor unit firing rate (Beck et al., 2007), presented a transient increase with time followed by a continuous reduction in both groups. In climbers, though, this behaviour seemed to be amplified towards higher values. This trend seems to suggest that in trained participants, motor units with a high force production capacity and high firing frequency are prevalently present. Moreover, similarly to EMG, the smaller skinfold thickness may have filtered the mechanical waves to a lesser extent in climbers. The comparison of the time and frequency domain parameters of the EMG and MMG signals after normalisation to their initial value at the beginning of the effort aimed to better distinguish possible specific trends in the two groups of participants. The dynamics of normalised RMS and MF through the exertion period in controls and climbers mostly overlap, suggesting that motor commands outflow, which determines the motor unit recruitment and firing rate interaction, is similar in the two groups. Overall, though, we need to make some considerations about the outcome of muscle functional changes in climbers from our results and data that are already present in the literature: (i) the presence of type I fibres larger than type II fibres (Farina, Merletti, & Enoka, 2004) and of specific motor units territory distributions with respect to the EMG detection site (Farina, Merletti, & Enoka, 2014) may provide an increase in global conduction velocity, and hence in EMG MF, independent of the prevalence of action potentials of fast motor units in the detected EMG; (ii) MMG MF is strongly influenced by the global motor unit firing frequency (Beck et al., 2007), and contrary to EMG MF, the higher values of MMG MF in climbers in the present study reached the statistical significance; (iii) type II fibre motor units present higher firing frequencies during sustained contractions (Burke, 1981; Freund, 1983; Grimby & Hannerz, 1977). Therefore, considering the longer time to exhaustion, the differences in EMG and MMG time and frequency domain parameters between climbers and controls, it seems to be reasonable to support the hypothesis that years of specific activity of the finger flexors may result in a larger functional representation, during muscle action, of fast and larger fibre motor units, with a lower index of fatigue, such as the fast resistant motor units are, even though EMG MF differences alone cannot play a pivotal role in supporting the hypothesis we made above.

Conclusions

In conclusion, a greater ability of climbers to involve type II fibre motor units in the force generation process has been previously observed, but a possible fast fatigable or fast resistant motor unit prevalence in elite climbers could not be assessed. In the present investigation, the higher MVC and specific tension, the longer endurance time, the better force stability and accuracy, the higher values and different behaviour of EMG and MMG parameters in climbers are all suggestive of a functional prevalence of large and less fatigable motor units, namely the fast resistant motor units (Schiaffino & Reggiani, 2011), rather than type II fast fatigable motor units. The combined EMG, MMG and force approach was a useful non-invasive means to get deeper insights into the type of motor unit more involved during a sustained contraction at 80% MVC in climbers. However, motor unit activation during specific climbers' grips requires further investigation. This methodological approach could reveal handy consequences also in other sport, work and clinical fields, especially when an invasive analysis cannot be performed. Future studies combining a similar approach with morphological findings through direct fibre type assessment by muscle biopsies would provide definitive evidence for the validity of the present methodology.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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